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Computational Methods for PDE

Mass and Energy preserving exponential Runge – Kutta methods for the Nonlinear Schrödinger equation

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Academic Year 2025-2026

Acknowledgments

We express our deepest gratitude to

Dr. Satyanarayana Chirala

Department of Mathematics
Mahindra University

(acknowledgement)

With sincere appreciation,
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Abstract

The nonlinear Schrödinger (NLS) equation is an important model in mathematical physics and appears in applications such as nonlinear optics, quantum mechanics, plasma physics, and wave propagation. One of the main difficulties in solving this equation numerically is maintaining its important physical properties, especially the conservation of mass and energy, over long time intervals. Many standard numerical methods may produce accurate short-term results, but they often accumulate errors and fail to preserve these quantities properly in long-time simulations.

This report focuses on exponential Runge–Kutta methods for the NLS equation, with special attention to schemes that preserve mass and energy. The equation is first reformulated by separating the stiff linear part from the nonlinear term, which makes it suitable for exponential integration. To improve the conservation properties of the numerical method, the Scalar Auxiliary Variable (SAV) approach is introduced. This reformulation converts the nonlinear energy into a form that is easier to handle while retaining the structure of the original problem. Combined with the Lawson transformation and symplectic Runge–Kutta time discretization, this leads to the Exponential Scalar Auxiliary Variable Runge–Kutta (ESAV-RK) scheme.

Overall, this study shows how structure-preserving exponential integrators can provide stable, accurate, and physically meaningful numerical solutions of the NLS equation, especially for long-time simulations where conservation properties are essential.

Contents

1 Introduction **5**

2 Model Reformulation **5**
2.1 Exponential Integrator Transformation 6
2.2 Spatial Discretisation 6
2.3 Advantages of Exponential Runge–Kutta Methods 6

3 Mass and Energy Preserving Schemes **7**
3.1 Introduction to Conservation Properties 7
3.2 The Conservation Laws of the NLS Equation 7
3.3 The Scalar Auxiliary Variable (SAV) Approach 8
3.4 The Lawson Transformation 8
3.5 Exponential Runge-Kutta Methods 9
3.6 The Complete ESAV-RK Scheme 10
3.7 Theoretical Guarantees 10
3.8 Spatial Discretization Considerations 11
3.9 Comparison with Other Approaches 11
3.10 Summary of Comparisons 12

4 Implementation, Results and Applications **12**
4.1 Practical Implementation Aspects 12

1 Introduction

The Nonlinear Schrödinger (NLS) equation is one of the most widely studied models in mathematical physics. It describes the evolution of a complex-valued wave function and appears in diverse areas such as nonlinear optics, plasma physics, deep water waves, Bose–Einstein condensates, and quantum mechanics. Because of its rich structure and long-time coherent wave behaviour, the NLS equation serves as an important benchmark for developing and analysing numerical methods.

The standard cubic NLS equation is given by

$$iu_t + u_{xx} + \lambda |u|^2 u = 0, \qquad x \in \Omega,\ t > 0, \tag{1}$$

where $u(x,t)$ is a complex-valued field and $\lambda = \pm 1$ denotes focusing ($\lambda = 1$) or defocusing ($\lambda = -1$) nonlinearity.

A key property of the NLS equation is the preservation of physical invariants. Two of the most important conserved quantities are:

- **Mass:**
$$M(u) = \int_\Omega |u(x,t)|^2 \, dx, \tag{2}$$
- **Energy:**
$$E(u) = \int_\Omega \left(|u_x(x,t)|^2 - \frac{\lambda}{2} |u(x,t)|^4 \right) dx. \tag{3}$$

When solving the NLS equation numerically, preserving these invariants is crucial for achieving physically meaningful long-time simulations. Standard schemes such as classical Runge–Kutta or Crank–Nicolson methods may suffer from error accumulation and fail to maintain mass and energy accurately over long time intervals.

To address these challenges, structure-preserving time integrators have been developed. Among them, *exponential Runge–Kutta (ERK) methods* are especially attractive because they treat the stiff linear part of the equation exactly via the matrix exponential, providing improved stability and accuracy for oscillatory problems. Recent research focuses on constructing ERK schemes that additionally preserve mass and energy at the discrete level, leading to highly accurate and physically consistent simulations.

2 Model Reformulation

The NLS equation can be written in operator form as

$$u_t = Lu + N(u), \tag{4}$$

where

$$Lu = iu_{xx}, \tag{5}$$
$$N(u) = -i\lambda |u|^2 u. \tag{6}$$

Here, L is a stiff linear differential operator responsible for rapid oscillations, whereas $N(u)$ is a nonlinear but non-stiff term. This splitting is a key step in developing exponential integrators, which integrate the linear part exactly while treating the nonlinear part explicitly.

2.1 Exponential Integrator Transformation

To remove stiffness associated with the linear operator L, we introduce the transformed variable

$$v(t) = e^{-tL}u(t). \tag{7}$$

Differentiating with respect to time gives

$$v_t = e^{-tL}N\big(e^{tL}v(t)\big)\,. \tag{8}$$

Thus, the problem is reformulated as the non-stiff ordinary differential equation

$$v_t = F(t, v) := e^{-tL}N\big(e^{tL}v\big)\,. \tag{9}$$

Once $v(t)$ is computed numerically, the original solution is recovered via

$$u(t) = e^{tL}v(t). \tag{10}$$

This transformation is the basis of exponential Runge–Kutta methods, which avoid the stiffness originating from the linear dispersive operator.

2.2 Spatial Discretisation

Let Ω be discretised using either Fourier spectral methods (for periodic boundaries) or finite differences. The continuous Laplacian operator u_{xx} is replaced by a discrete operator D_{xx}, so the semi-discrete system takes the form

$$\frac{d}{dt}u(t) = Lu(t) + N\big(u(t)\big), \qquad L = iD_{xx}. \tag{11}$$

An exponential integrator then computes the numerical solution using

$$u^{n+1} = e^{hL}u^n + \text{nonlinear correction terms}, \tag{12}$$

where h is the time step size.

2.3 Advantages of Exponential Runge–Kutta Methods

Exponential Runge–Kutta schemes offer several key advantages:

- Exact treatment of the stiff linear part via matrix exponentials.
- Ability to use larger time steps without losing stability.
- High accuracy for oscillatory and dispersive wave equations.
- Natural compatibility with structure-preserving constraints.
- Possibility of preserving mass and energy at the discrete level.

These properties make ERK methods particularly suitable for long-time simulations of the NLS equation and other nonlinear dispersive PDEs.

3 Mass and Energy Preserving Schemes

3.1 Introduction to Conservation Properties

One of the most remarkable features of the NLS equation is its ability to preserve certain physical quantities over time, namely mass and energy. These conservation laws are not merely mathematical curiosities; they represent fundamental physical principles that any reliable numerical method should respect.

When we solve the NLS equation numerically, we face a critical challenge: how do we ensure that our discrete numerical schemes preserve these essential conservation properties? Standard numerical methods, even highly accurate ones, often fail to maintain exact conservation of mass and energy.

Structure-preserving numerical methods address this challenge by maintaining the geometric and algebraic properties of the continuous system at the discrete level. This report explores how combining the Scalar Auxiliary Variable (SAV) approach with exponential Runge-Kutta methods achieves exact mass and energy preservation.

3.2 The Conservation Laws of the NLS Equation

Before diving into the numerical schemes, let us establish what we mean by mass and energy conservation for the NLS equation. Consider the cubic nonlinear Schrödinger equation:

$$i\frac{\partial \psi}{\partial t} = -\frac{1}{2}\Delta\psi + \beta|\psi|^2\psi \tag{13}$$

where $\psi(x, y, t)$ is the complex-valued wave function, $i = \sqrt{-1}$ is the imaginary unit, Δ is the Laplacian operator, and β is a real parameter governing the strength of the nonlinearity.

This equation preserves two fundamental quantities. The **mass** (sometimes called the L^2 norm or charge) is defined as:

$$M(t) = \int_\Omega |\psi(x, y, t)|^2 \, dx \, dy \tag{14}$$

This quantity represents the total probability in quantum mechanics or the total power in optical applications. The mass conservation law states that $M(t) = M(0)$ for all time $t \geq 0$.

The **energy** (or Hamiltonian) is given by:

$$H(t) = \int_\Omega \left[\frac{1}{2}|\nabla\psi|^2 + \frac{\beta}{2}|\psi|^4\right] dx \, dy \tag{15}$$

This functional consists of two parts: the kinetic energy (gradient term) and the potential energy (nonlinear term). The energy conservation law states that $H(t) = H(0)$ for all time.

These conservation laws can be verified by direct differentiation and application of the equation itself. For mass conservation, we compute:

$$\frac{d}{dt}M(t) = \int_\Omega \frac{\partial}{\partial t}|\psi|^2 \, dx \, dy = \int_\Omega \left(\frac{\partial \psi}{\partial t}\overline{\psi} + \psi\frac{\partial \overline{\psi}}{\partial t}\right) dx \, dy \tag{16}$$

Substituting the NLS equation and using integration by parts with periodic boundary conditions, one can show that this derivative vanishes, proving mass conservation.

3.3 The Scalar Auxiliary Variable (SAV) Approach

The key innovation in developing mass and energy preserving schemes lies in the SAV approach, introduced by Shen, Xu, and Yang in 2018 [1]. This technique has revolutionized the design of energy-stable schemes for gradient flows and Hamiltonian systems. The fundamental idea is to reformulate the original equation by introducing an auxiliary scalar variable that encapsulates the nonlinear energy contribution.

For the NLS equation, the SAV reformulation proceeds as follows. We introduce an auxiliary variable:

$$q(t) = \sqrt{(\psi^2, \psi^2) + C_0} \tag{17}$$

where $(f, g) = \int_\Omega f\overline{g}\, dx\, dy$ denotes the L^2 inner product, and $C_0 > 0$ is a stabilization constant chosen to ensure that q remains well-defined and bounded away from zero for all time. The choice of C_0 is problem-dependent but typically set to ensure $(\psi^2, \psi^2) + C_0 > 0$ throughout the simulation.

The original energy functional can now be rewritten in a quadratic form:

$$E(t) = (L\psi, \psi) + \frac{\beta}{2}q^2 - \frac{\beta}{2}C_0 \tag{18}$$

where $L\psi = -\frac{1}{2}\Delta\psi$ represents the linear operator. This reformulation is crucial because it transforms the quartic nonlinearity $|\psi|^4$ into a quadratic term q^2, which is much easier to handle in numerical schemes.

Using the energy variational principle, the original NLS equation can be equivalently written as the following SAV reformulated system:

$$\frac{\partial \psi}{\partial t} = -i\left(L\psi + \frac{\beta|\psi|^2\psi \cdot q}{\sqrt{(\psi^2, \psi^2) + C_0}}\right) \tag{19}$$

$$\frac{dq}{dt} = 2\mathrm{Re}\left(\frac{\partial \psi}{\partial t}, \frac{|\psi|^2\psi}{\sqrt{(\psi^2, \psi^2) + C_0}}\right) \tag{20}$$

This reformulated system is mathematically equivalent to the original NLS equation at the continuous level, but it explicitly separates the stiff linear part from the nonlinear part, making it amenable to exponential integrator techniques.

3.4 The Lawson Transformation

The next key step in developing our structure-preserving scheme is the Lawson transformation, a classical technique introduced by J.D. Lawson in 1967 [2]. As an exponential integrator, it is designed to handle stiff systems efficiently and has gained renewed interest for its stability and ability to deal with highly oscillatory dynamics.

The core idea is straightforward: separate the stiff linear part from the nonlinear part, integrate the linear component exactly, and apply a standard time-stepping method to the remaining nonlinear term. For the NLS equation, this is especially useful because the linear Schrödinger operator induces fast oscillations that would otherwise require extremely small time steps with traditional methods.

We define the change of variables:

$$u(x, y, t) = \exp(iLt)\psi(x, y, t) \tag{21}$$

This transformation essentially filters out the oscillations generated by the linear operator L. In the new variable u, the system becomes:

$$\frac{\partial u}{\partial t} = -i\exp(iLt)\frac{\beta|\exp(-iLt)u|^2\exp(-iLt)u\cdot q}{\sqrt{((\exp(-iLt)u)^2,(\exp(-iLt)u)^2)+C_0}} \tag{22}$$

$$\frac{dq}{dt} = 2\mathrm{Re}\left(-i\exp(-iLt)Lu, \frac{|\exp(-iLt)u|^2\exp(-iLt)u}{\sqrt{((\exp(-iLt)u)^2,(\exp(-iLt)u)^2)+C_0}}\right) \tag{23}$$

While this system may appear more complex, it has transformed the problem into one where the fast oscillations are eliminated, and the remaining dynamics are much smoother. This allows for the use of larger time steps while maintaining accuracy and stability.

Importantly, the Lawson-transformed system preserves modified conservation laws that are equivalent to the original ones:

$$\tilde{M}(t) = (u,u) = M(0) \tag{24}$$

$$\tilde{E}(t) = (Lu,u) + \frac{\beta}{2}q^2 - \frac{\beta}{2}C_0 = E(0) \tag{25}$$

3.5 Exponential Runge-Kutta Methods

With the SAV-Lawson reformulation in hand, we now turn to the temporal discretization using Runge-Kutta methods. Runge-Kutta methods are among the most widely used time-stepping schemes in scientific computing, known for their high order accuracy and flexibility. However, not all Runge-Kutta methods preserve energy—we need to carefully select methods that satisfy certain algebraic conditions.

A general s-stage Runge-Kutta method is characterized by its Butcher tableau, consisting of coefficients a_{ij} (for $i,j=1,\ldots,s$), weights b_i, and nodes $c_i = \sum_{j=1}^{s} a_{ij}$. Applied to the Lawson-transformed system, the method computes intermediate stages:

$$U_i = u^n + \tau\sum_{j=1}^{s} a_{ij}\tilde{k}_j, \quad Q_i = q^n + \tau\sum_{j=1}^{s} a_{ij}l_j \tag{26}$$

where τ is the time step, and $\tilde{k}_i$ and l_i represent the stage derivatives evaluated at the intermediate points.

The crucial observation, proven by Hairer, Lubich, and Wanner in their seminal textbook on geometric numerical integration [3], is that Runge-Kutta methods preserve quadratic invariants if and only if their coefficients satisfy the symplecticity condition:

$$b_i a_{ij} + b_j a_{ji} = b_i b_j, \quad \forall i,j = 1,\ldots,s \tag{27}$$

This condition characterizes symplectic Runge-Kutta methods. When applied to Hamiltonian systems (including our reformulated NLS equation), symplectic methods preserve the symplectic structure and, consequently, the energy.

3.6 The Complete ESAV-RK Scheme

Combining all the ingredients—SAV reformulation, Lawson transformation, and symplectic Runge-Kutta methods—we arrive at the Exponential Scalar Auxiliary Variable Runge-Kutta (ESAV-RK) method. After carefully manipulating the exponentials to return to the original variable ψ, the complete scheme takes the form:

Stage computations:

$$\Psi_i = \exp(-iLc_i\tau)\psi^n + \tau \sum_{j=1}^{s} a_{ij} \exp(iL(c_j - c_i)\tau)k_j \tag{28}$$

$$Q_i = q^n + \tau \sum_{j=1}^{s} a_{ij} l_j \tag{29}$$

where the stage derivatives are:

$$k_i = -i \frac{\beta|\Psi_i|^2\Psi_i \cdot Q_i}{\sqrt{(\Psi_i^2, \Psi_i^2) + C_0}} \tag{30}$$

$$l_i = 2\mathrm{Re}\left(-iL\Psi_i, \frac{|\Psi_i|^2\Psi_i}{\sqrt{(\Psi_i^2, \Psi_i^2) + C_0}}\right) \tag{31}$$

Solution update:

$$\psi^{n+1} = \exp(-iL\tau)\psi^n + \tau \sum_{i=1}^{s} b_i \exp(-iL(1 - c_i)\tau)k_i \tag{32}$$

$$q^{n+1} = q^n + \tau \sum_{i=1}^{s} b_i l_i \tag{33}$$

This scheme is remarkably elegant: it integrates the linear part exactly using matrix exponentials, treats the nonlinear part with a high-order Runge-Kutta method, and maintains the SAV structure that ensures energy preservation.

3.7 Theoretical Guarantees

The theoretical foundation of the ESAV-RK method rests on several key lemmas and theorems. A fundamental property of the exponential operator is established through the following result:

For the symmetric positive definite operator L and the operator $\exp(iLt)$:

1. $\exp(iLt)$ commutes with L
2. $\exp(iLt)^* = \exp(-iLt)$, where $*$ denotes the adjoint operator

These properties are crucial for proving conservation. They ensure that the exponential propagator preserves the inner product structure and commutes with the spatial operators.

The main conservation theorem can now be stated:

If the Runge-Kutta coefficients satisfy the symplecticity condition $b_i a_{ij} + b_j a_{ji} = b_i b_j$ for all i, j, then the ESAV-RK method exactly preserves the discrete mass and modified energy:

$$M^n = (\psi^n, \psi^n) = M^0 \tag{34}$$

$$E^n = (L\psi^n, \psi^n) + \frac{\beta}{2}(q^n)^2 - \frac{\beta}{2}C_0 = E^0 \tag{35}$$

for all time steps $n = 1, 2, \ldots, N$.

The proof relies on the fact that symplectic Runge-Kutta methods preserve quadratic first integrals of the underlying differential equation. By reformulating the energy as a quadratic invariant through the SAV approach, we bring it into the class of quantities automatically preserved by symplectic methods.

3.8 Spatial Discretization Considerations

While our focus has been on temporal discretization, achieving full conservation in practice requires careful attention to spatial discretization as well. The spatial discretization must preserve three key properties:

1. **Symmetric positive definiteness** of the operator L
2. **Discrete integration by parts** formulas
3. **High-order accuracy** compatible with the temporal method

For periodic boundary conditions, the Fourier pseudo-spectral method is the natural choice. This method represents the solution as a truncated Fourier series and computes derivatives by multiplication in Fourier space. It satisfies all three requirements above and provides spectral accuracy when the solution is smooth.

The discrete Laplacian in Fourier space becomes simply:

$$\widehat{L\psi}_{\mathbf{k}} = \frac{1}{2}|\mathbf{k}|^2 \widehat{\psi}_{\mathbf{k}} \tag{36}$$

where $\mathbf{k}$ represents the wave vector and $\widehat{\cdot}$ denotes the Fourier transform. This diagonal structure in Fourier space makes the computation of matrix exponentials like $\exp(-iL\tau)$ extremely efficient—it reduces to pointwise multiplication by $\exp(-i|\mathbf{k}|^2\tau/2)$.

3.9 Comparison with Other Approaches

The ESAV-RK method can be compared with several other structure-preserving approaches in the literature:

Crank-Nicolson schemes: These classical second-order methods are widely used for the NLS equation and preserve mass exactly. However, they only preserve energy approximately (with second-order accuracy) and are limited to second-order temporal accuracy. The ESAV-RK methods achieve arbitrary high order while preserving both mass and energy exactly.

Discrete variational derivative methods: Introduced by Furihata and Matsuo [4], these methods are based on discretizing the variational structure of the equation. While they preserve a discrete analogue of energy, they typically require solving nonlinear systems at each time step and can be computationally expensive for high-order schemes.

Symmetric exponential integrators: Celledoni, Cohen, and Owren [5] developed symmetric exponential integrators that preserve energy through projection. However, projection methods can be expensive and may not preserve mass simultaneously.

The ESAV-RK approach offers several advantages: it preserves both mass and energy simultaneously without projection, achieves arbitrary high order through choice of Runge-Kutta method, and benefits from the excellent stability properties of exponential integrators for stiff and oscillatory problems.

3.10 Summary of Comparisons

The ESAV-RK method can be compared systematically with other structure-preserving approaches. Table 1 summarizes the key properties of different methods.

Table 1: Comparison of structure-preserving methods for the nonlinear Schrödinger equation

Method	**Mass Conservation**	**Energy Conservation**	**Order**	**Projection**	**Nonlinear Solve**
Crank-Nicolson	Exact	$O(\tau^2)$	2	No	Yes
Discrete Variational	Exact	Exact	Variable	No	Yes (complex)
Sym, Exp. + Proj.	Depends	Exact	High	Yes	Depends
ESAV-RK	**Exact**	**Exact**	**Arbitrary**	**No**	**Yes (simpler)**

The ESAV-RK method offers the best combination: exact conservation of both mass and energy, arbitrary high order, no projection, and relatively simple nonlinear systems due to the SAV reformulation.

4 Implementation, Results and Applications

4.1 Practical Implementation Aspects

Implementing the ESAV-RK method requires careful attention to several computational details. The evaluation of matrix exponentials $\exp(-iL\tau)$ is performed efficiently in Fourier space using FFTs. For a 2D problem with $N \times N$ grid points, this requires $O(N^2 \log N)$ operations.

The choice of stabilization constant C_0 deserves consideration. In practice, C_0 is typically chosen based on the initial condition: $C_0 = \epsilon \cdot (\psi_0^2, \psi_0^2)$ where $\epsilon \in [0.01, 0.1]$ is a small parameter. This ensures that the denominator in the SAV formulation remains well-conditioned throughout the simulation.

The stage computations can be performed in parallel for explicit Runge-Kutta methods, though the high-order Gauss methods used in the paper (ESAV-RK4 and ESAV-RK6) are implicit and require iterative solvers for the stage equations. However, the SAV reformulation linearizes much of the nonlinearity, making these iterations converge rapidly.

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